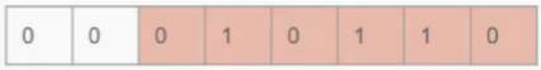
Binary Left and Right Shift

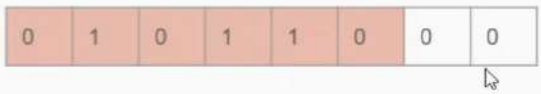
Bit Manipulation Lecture : 01

</ CodeNCode >

Binary Left Shift

N = 22



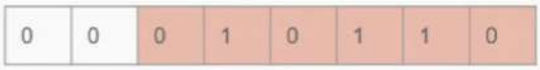


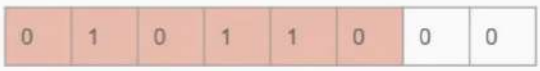
N = N << 2;

</ CodeNCode >

Binary Left Shift

N = 22



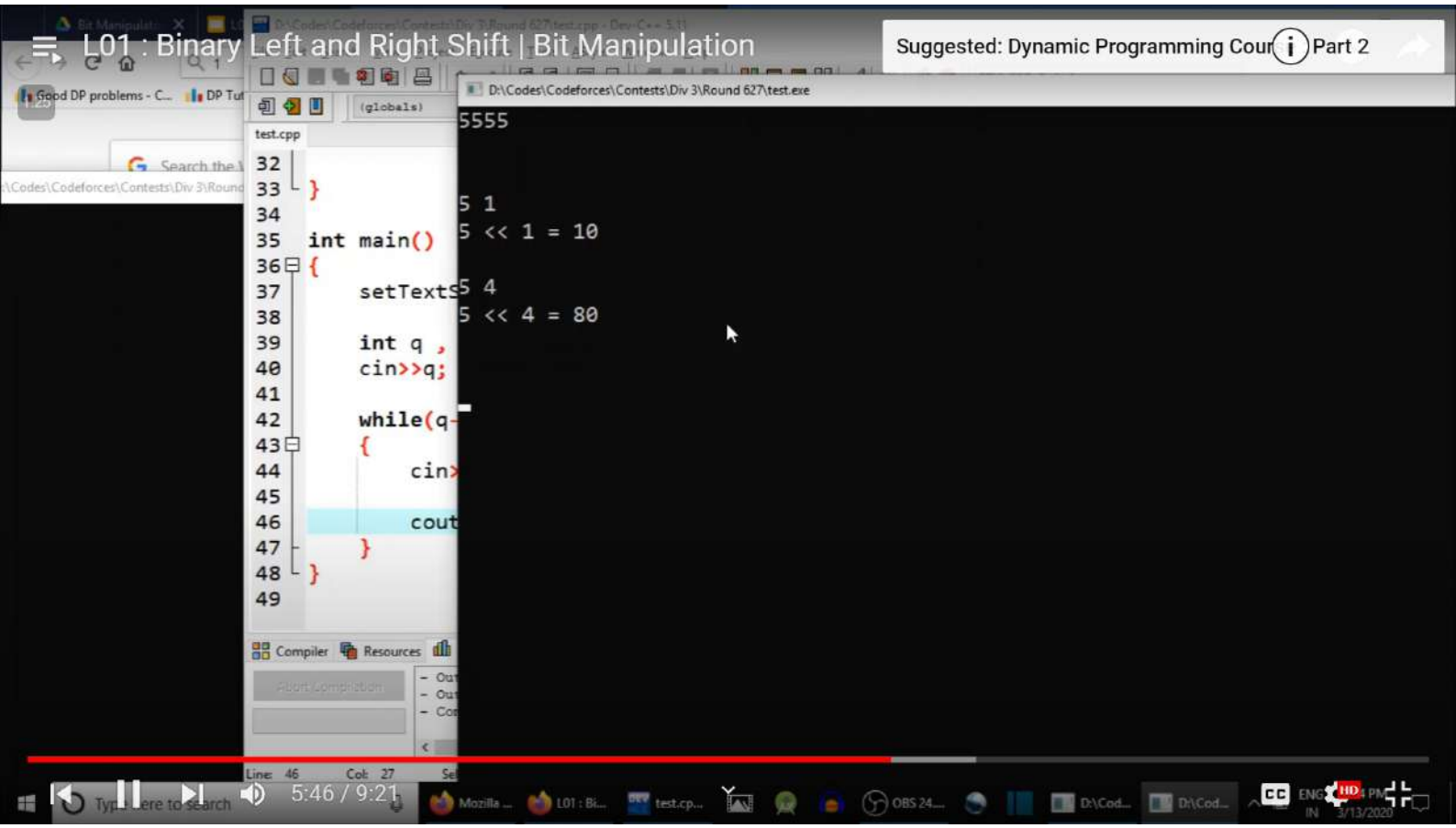


N = N << 2;

N = N << i;

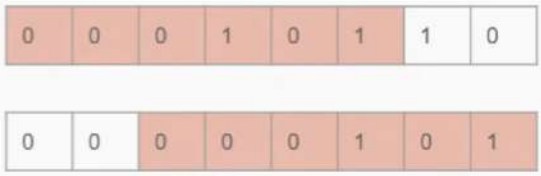
N = N * (2^i)

</ CodeNCode >


L01 : Binary Left and Right Shift | Bit Manipulation
Suggested: Dynamic Programming Cour
Part 2
D:\Codes\Codeforces\Contests\Div 3\Round 627\test.exe
5555
5 1
5 << 1 = 10
5 4
5 << 4 = 80
test.cpp
int main()
{
int q ,
cin>>q;
while(q-
{
cout
}
}
Compiler
Resources
5:46 / 9:21

Binary Left Shift

N = 22



N = N >> 2;

N = N >> i;

N = N / (2^i)

</ CodeNCode >

Checking for ith set bit

Bit Manipulation Lecture : 02

</ CodeNCode >

Problem Statement

Given a number N and index i , find whether ith bit in binary representation of N is set or not

Example :

N = 12 (1100)	i = 0	ans = false
N = 12 (1100)	i = 3	ans = true

</ CodeNCode >

Checking for ith set bit

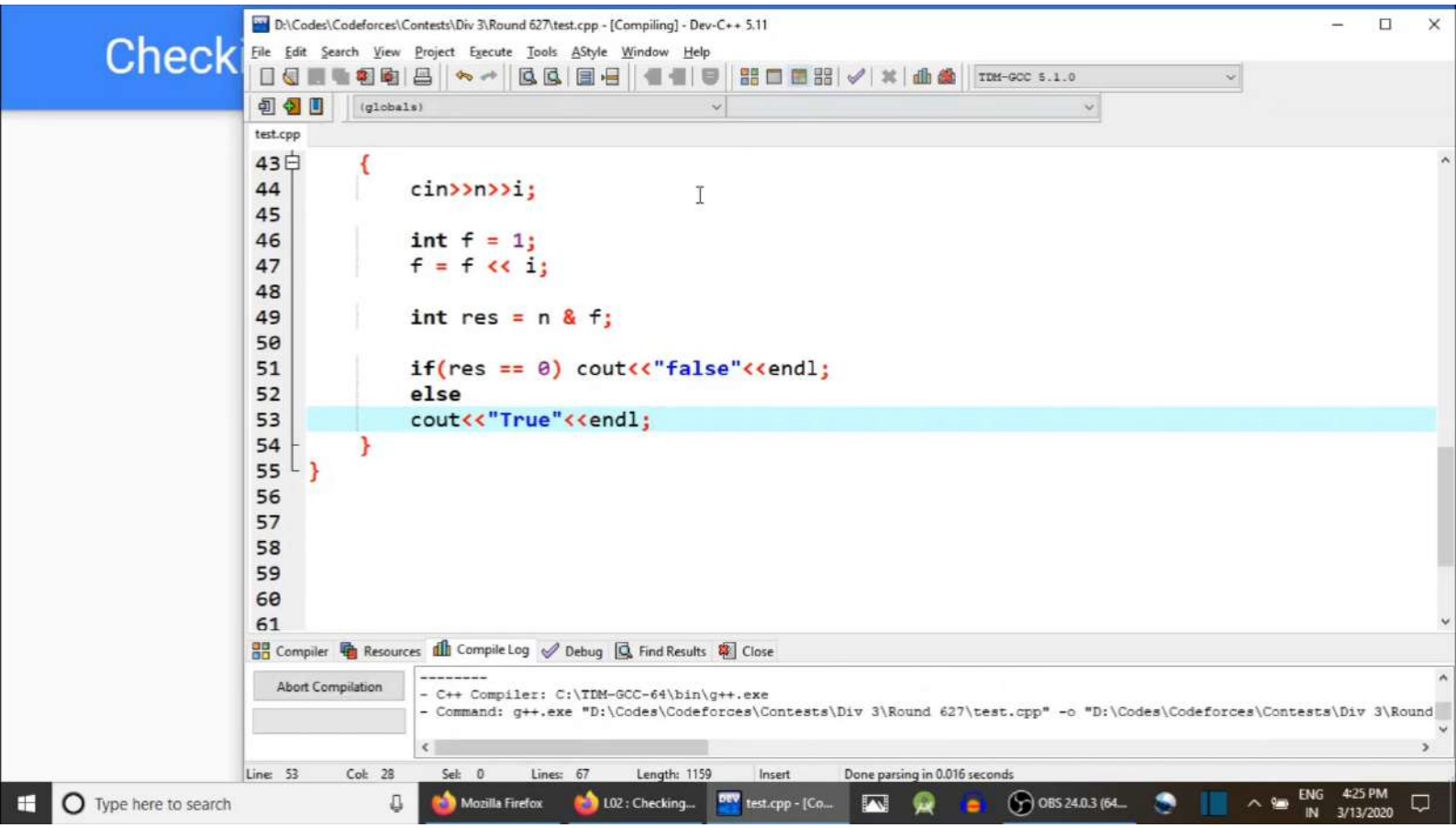
N = 12
00001100

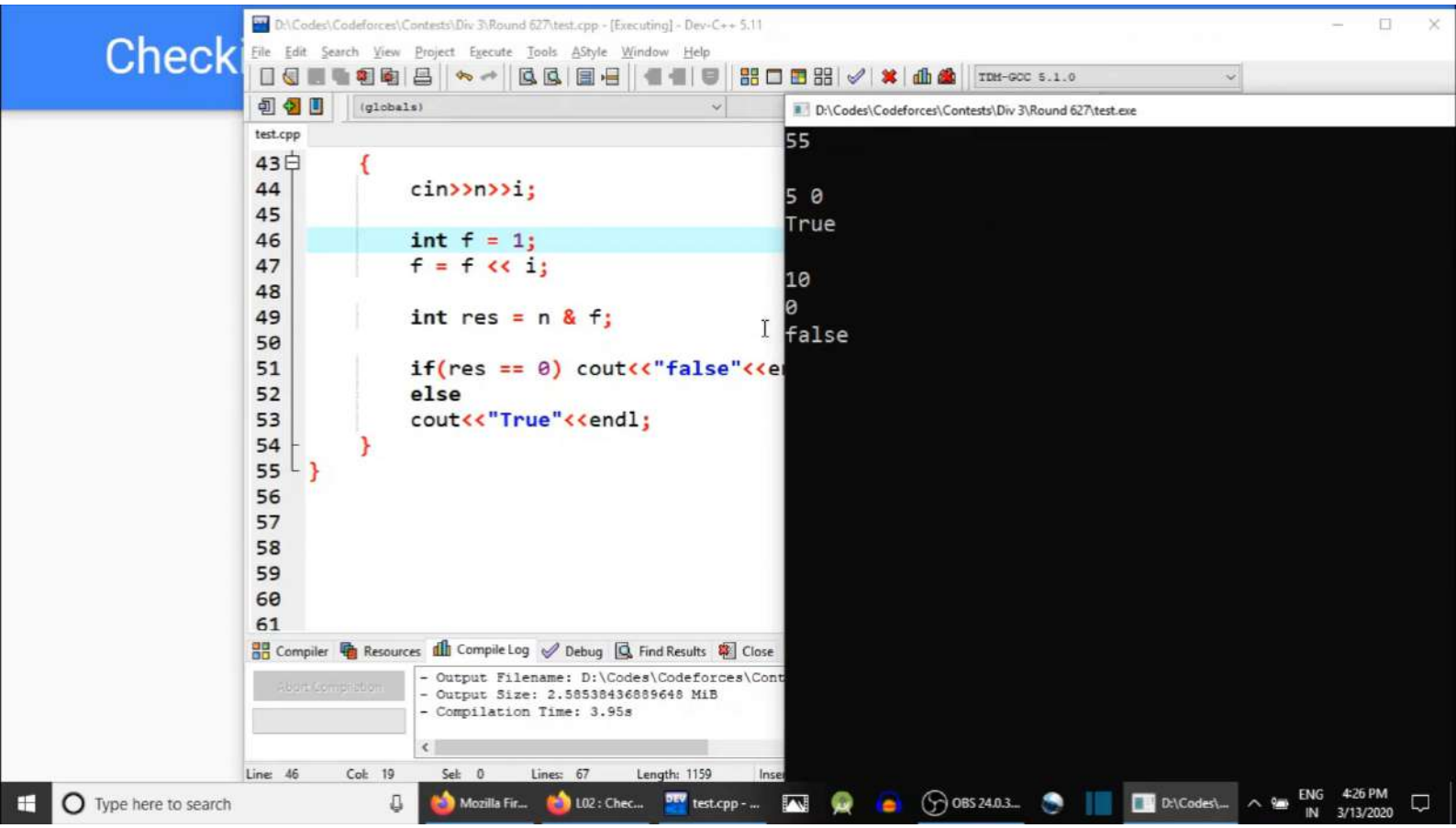
F = 1
00000001

F = F << 3;
00001000

```
00001100          00001100
00001000          00010000
----------        ----------
00001000          00000000
```

</ CodeNCode >


Check
D:\Codes\Codeforces\Contests\Div 3\Round 627\test.cpp - [Compiling] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 5.1.0
(globals)
test.cpp
{
cin>>n>>i;
int f = 1;
f = f << i;
int res = n & f;
if(res == 0) cout<<"false"<<endl;
else
cout<<"True"<<endl;
}
}
Compiler Resources Compile Log Debug Find Results Close
Abort Compilation
- C++ Compiler: C:\TDM-GCC-64\bin\g++.exe
- Command: g++.exe "D:\Codes\Codeforces\Contests\Div 3\Round 627\test.cpp" -o "D:\Codes\Codeforces\Contests\Div 3\Round
Line: 53 Col: 28 Sel: 0 Lines: 67 Length: 1159 Insert Done parsing in 0.016 seconds
Type here to search
Mozilla Firefox
L02 : Checking...
test.cpp - [Co...
OBS 24.0.3 (64...
ENG
IN
4:25 PM
3/13/2020


Check
D:\Codes\Codeforces\Contests\Div 3\Round 627\test.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 5.1.0
(globals)
test.cpp
43 {
44 cin>>n>>i;
45
46 int f = 1;
47 f = f << i;
48
49 int res = n & f;
50
51 if(res == 0) cout<<"false"<<e
52 else
53 cout<<"True"<<endl;
54 }
55 }
56
57
58
59
60
61
Compiler Resources Compile Log Debug Find Results Close
Abort Compilation
- Output Filename: D:\Codes\Codeforces\Cont
- Output Size: 2.58538436889648 MiB
- Compilation Time: 3.95s
Line: 46 Col: 19 Sel: 0 Lines: 67 Length: 1159
D:\Codes\Codeforces\Contests\Div 3\Round 627\test.exe
55
5 0
True
10
0
false
Type here to search
Mozilla Fir...
L02 : Chec...
test.cpp - ...
OBS 24.0.3...
D:\Codes\...
ENG
IN
4:26 PM
3/13/2020

Counting Number of set bits

Bit Manipulation Lecture : 03

</ CodeNCode >

Problem Statement

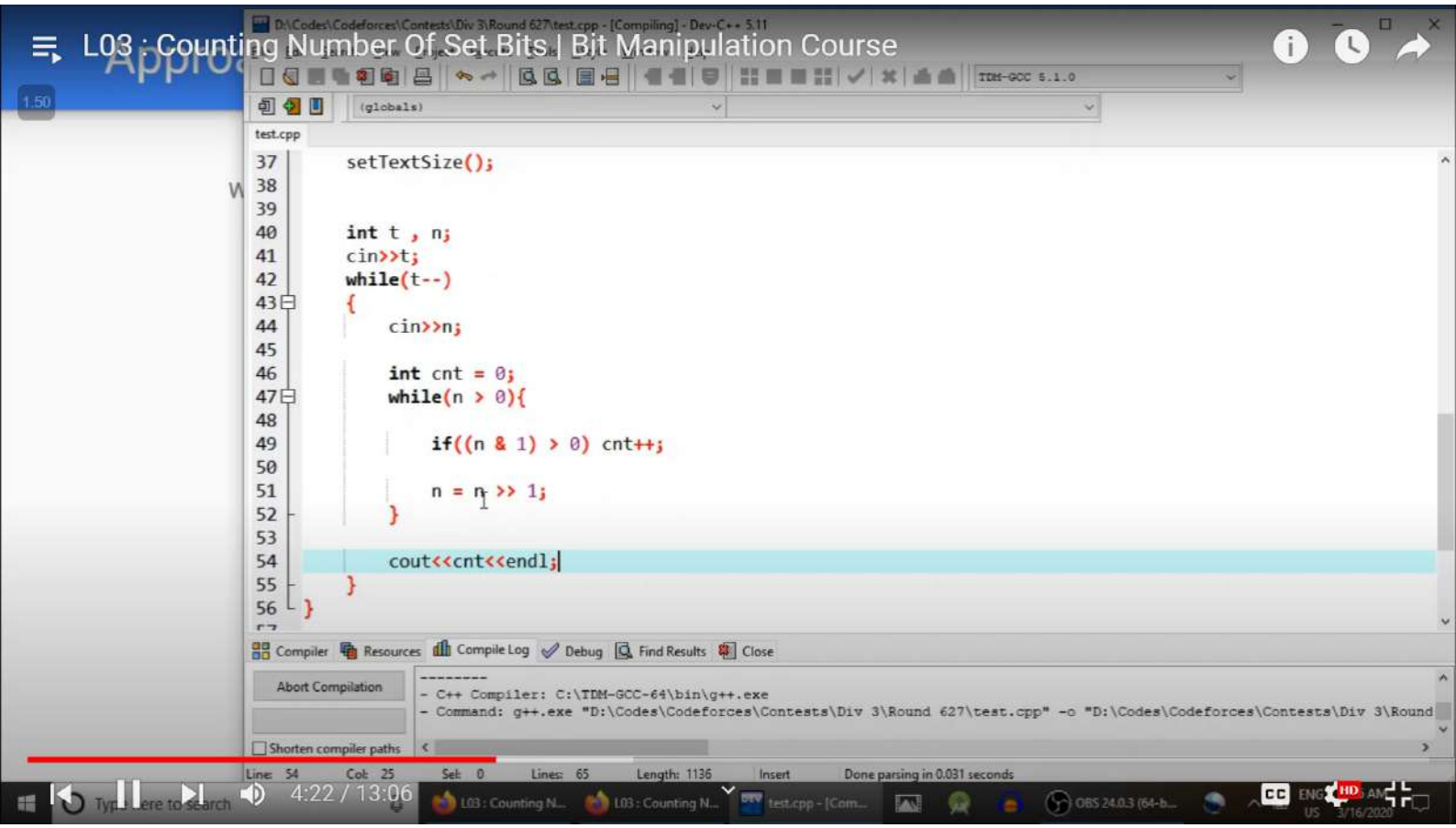
Given a number N and find number of set bits in it.

Example :

N = 12 (1100) output : 2

N = 11 (1011) output : 3

</ CodeNCode >


L03 : Counting Number Of Set Bits | Bit Manipulation Course
1.50
D:\Codes\Codeforces\Contests\Div 3\Round 627\test.cpp - [Compiling] - Dev-C++ 5.11
TDM-GCC 5.1.0
(globals)
test.cpp
setTextSize();
int t , n;
cin>>t;
while(t--)
{
cin>>n;
int cnt = 0;
while(n > 0){
if((n & 1) > 0) cnt++;
n = n >> 1;
}
cout<<cnt<<endl;
}
}
Compiler
Resources
Compile Log
Debug
Find Results
Close
Abort Compilation
Shorten compiler paths
- C++ Compiler: C:\TDM-GCC-64\bin\g++.exe
- Command: g++.exe "D:\Codes\Codeforces\Contests\Div 3\Round 627\test.cpp" -o "D:\Codes\Codeforces\Contests\Div 3\Round
Line: 54 Col: 25 Sel: 0 Lines: 65 Length: 1136 Insert Done parsing in 0.031 seconds
4:22 / 13:06

Another approach

Another approach is based on a simple observation

8 - 1 = 7

```
01000
00001
--------
00111
```

16 - 1 = 15

```
10000
00001
--------
01111
```

28 - 1 = 27

```
11100
00001
--------
11011
```

</ CodeNCode >

Another approach

```
int cnt = 0;

while(n > 0)
{
    cnt++;

   n = n & (n-1);

}

return cnt;
```

```
0 0 1 0 1 1 0 1
```

```
0 0 1 0 1 1 0 1
0 0 1 0 1 1 0 0
---------------
0 0 1 0 1 1 0 0
```

</ CodeNCode >

Another approach

```
int cnt = 0;

while(n > 0)
{
    cnt++;

   n = n & (n-1);

}

return cnt;
```

```
00101100
```

```
00101100
00101011
------------
00101000
```

</ CodeNCode >

Another approach

```
int cnt = 0;

while(n > 0)
{
    cnt++;

   n = n & (n-1);

}

return cnt;
```

```
00101000
```

```
00101000
00100111
--------
00100000
```

</ CodeNCode >

Another approach

```
int cnt = 0;

while(n > 0)
{
    cnt++;

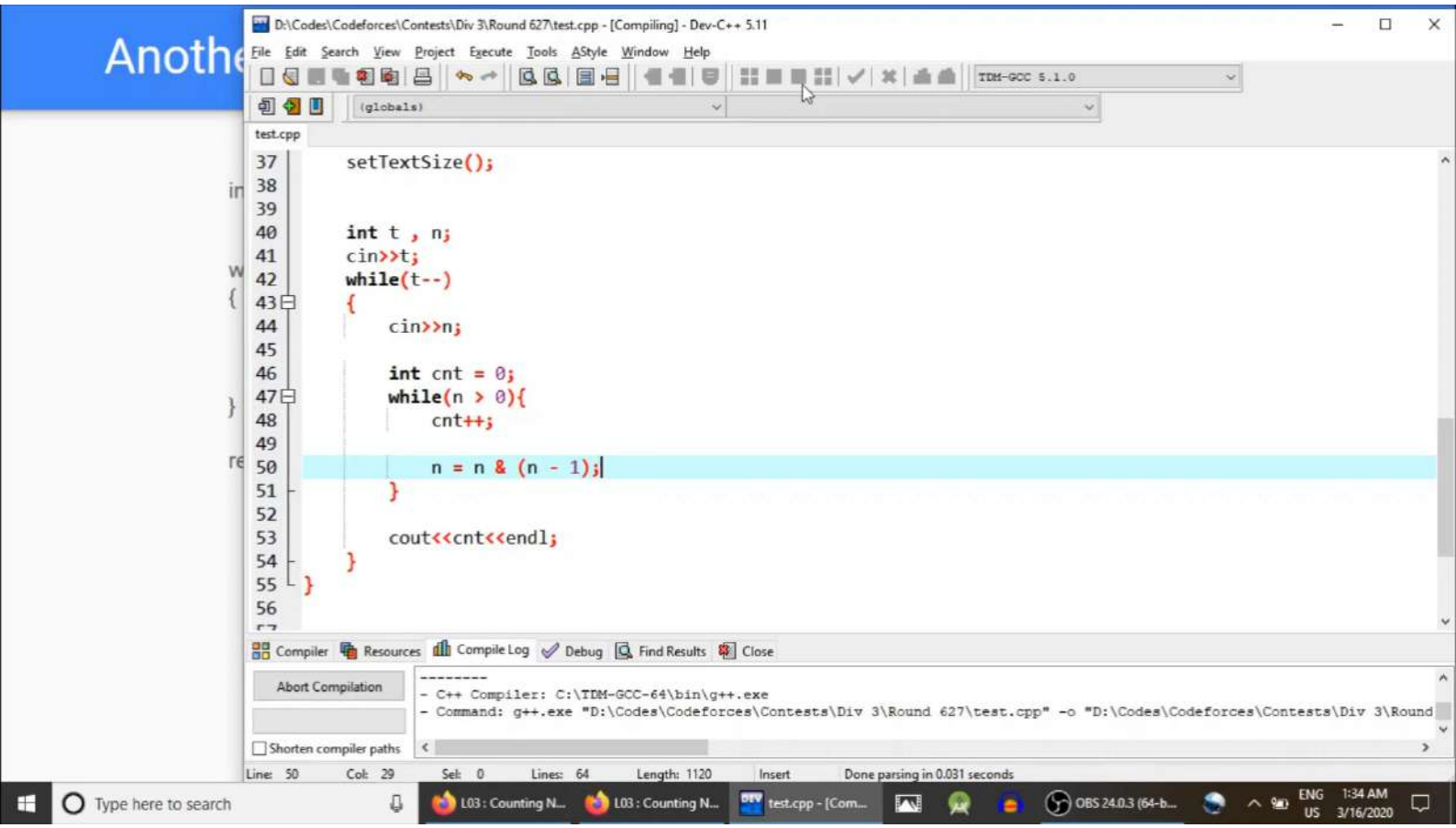
   n = n & (n-1);
}

return cnt;
```

```
00100000
```

```
00100000
00011111
---------
00000000
```

</ CodeNCode >


Anothe
D:\Codes\Codeforces\Contests\Div 3\Round 627\test.cpp - [Compiling] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 5.1.0
(globals)
test.cpp
37 setTextSize();
38
39
40 int t , n;
41 cin>>t;
42 while(t--)
43 {
44 cin>>n;
45
46 int cnt = 0;
47 while(n > 0){
48 cnt++;
49
50 n = n & (n - 1);
51 }
52
53 cout<<cnt<<endl;
54 }
55 }
56
Compiler Resources Compile Log Debug Find Results Close
Abort Compilation
Shorten compiler paths

- C++ Compiler: C:\TDM-GCC-64\bin\g++.exe
- Command: g++.exe "D:\Codes\Codeforces\Contests\Div 3\Round 627\test.cpp" -o "D:\Codes\Codeforces\Contests\Div 3\Round
Line: 50 Col: 29 Sel: 0 Lines: 64 Length: 1120 Insert Done parsing in 0.031 seconds
Type here to search
L03 : Counting N...
L03 : Counting N...
test.cpp - [Com...
OBS 24.0.3 (64-b...
ENG US
1:34 AM 3/16/2020

XOR and its properties

Bit Manipulation Lecture : 04

</ CodeNCode >

Bitwise XOR Operation

What is bitwise XOR Operation ?

Example :

A = 011001010
B = 010010110

A XOR B = 001011100

</ CodeNCode >

Bitwise XOR Operation

What is bitwise XOR Operation ?

Example :

A = 011001010
B = 010010110

A XOR B = 001011100

A XOR B = A ^ B

</ CodeNCode >

Properties of XOR

1. Identity element = 0

   0 ^ A = A

   A = 23 = 1 0 1 1 1

   A ^ 0 = 1 0 1 1 1
           0 0 0 0 0
           ----
           1 0 1 1 1

</ CodeNCode >

Properties of XOR

2. A ^ A = 0

A = 23 = 1 0 1 1 1
A ^ A = 1 0 1 1 1
1 0 1 1 1
---
0 0 0 0 0

</ CodeNCode >

Properties of XOR

3. Ordering of XOR doesn't matter

$$
\begin{aligned}
2 \wedge 3 \wedge 1 &= (2 \wedge 3) \wedge 1 \\
&= 2 \wedge (1 \wedge 3) \\
&= 3 \wedge (1 \wedge 2)
\end{aligned}
$$

</ CodeNCode >

Problem Statement

Given an array with n elements , find missing and repeating number in the array

1 <= Ai <= N , all integers are unique , except the repeating number which appears exactly twice.

N = 7
A[ ] = [1 , 3 , 2 , 5 , 5 , 6 , 7]

Missing Number : 4
Repeating Number : 5

</ CodeNCode >

Bit Manipulation

Power of 2
1 << n = 2 ^ n

1 << 0 = (1) = 2 ^ 0
1 << 1 = (1 0) = 2 ^ 1

*Bit :(
ith bot of x is 1 or not :
x & (1 << i) = 0 : ith bit of x is not 1
x & (1 << i) != 0 : ith bit of x is 1



Power Set

0 → x
1 → ✓

0 1 2

S = "abc" ⟶ "", a, b, c, ab, ac, bc, abc

num ↓	2	1	0 ← ind	
0	0	0	0	→ ""
1	0	0	1	→ a
2	0	1	0	→ b
3	0	1	1	→ ab
4	1	0	0	→ c
5	1	0	1	→ ac
6	1	1	0	→ bc
7	1	1	1	→ abc

TUF

```cpp
#include<bits/stdc++.h>
using namespace std;


vector<vector<int> > getSubsets(vector<int> num){
    int size = num.size();    // size = 3 num = {1,2,3}
    int subsetNum = (1<<size); /// subsetNum = 8

    vector<vector<int> > allSubsets;
    for(int subsetMask = 0; subsetMask < subsetNum; ++subsetMask){
        vector<int> subset;
        for(int i = 0; i < size; ++i){
            if ((subsetMask & (1 << i)) != 0){
                subset.push_back(num[i]);
            }
        }
        allSubsets.push_back(subset);
    }
    return allSubsets;
}


int main(){

}
```

Binary of numbers
{1,2,3}
0 = 000 = {}
1 = 001 ={3}
2 = 010 = {2}
3 = 011 = {2,3}
4 = 100 = {1}
5 = 101 = {1,3}
6 = 110 = {1,2}
7 = 111 = {1,2,3}
8 = 1000

